PROTOTYPE AI RESUME SCREENING TOOL

BAIM 660 -01

ALBRIGHT ARTHUR
JONATHAN MORRIS
ROHAN ALLE

Flow of the Project

Phase 1 :

- Clean data
- Train Test Split
- Encoding and Vectorization
- Model Training
- Resume Prediction Code

Phase 2 :

- Fix issues with Phase 1
- Deploy on Web Application
- Test with Text and PDF resumes

Phase 1Updates

- Logic for Resume Screening
- Fixing Split and Vectorization - split the dataset first, then vectorize
- Changed model from SVC to Linear SVC.

DATASET SUMMARY

- **Category** — The target label describing the type of resume
- **Resume** — The full text content of each resume, often cleaned and preprocessed (though still containing varied text formats).
- It typically has around **960 resumes across 25–30** unique categories, such as:
- Data Science, HR, Sales, Testing, Operations, Networking, Arts, Health, Finance, Mechanical, etc.

Category	count
Java Developer	84
Testing	70
DevOps Engineer	55
Python Developer	48
Web Designing	45
HR	44
Hadoop	42
Sales	40
Data Science	40
Mechanical Engineer	40
ETL Developer	40
Blockchain	40
Operations Manager	40
Arts	36
Database	33
Health and fitness	30
PMO	30
Electrical Engineering	30
Business Analyst	28
DotNet Developer	28
Automation Testing	26
Network Security Engineer	25
Civil Engineer	24
SAP Developer	24
Advocate	20

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Model Training: switch from SVC to LINEAR SVC

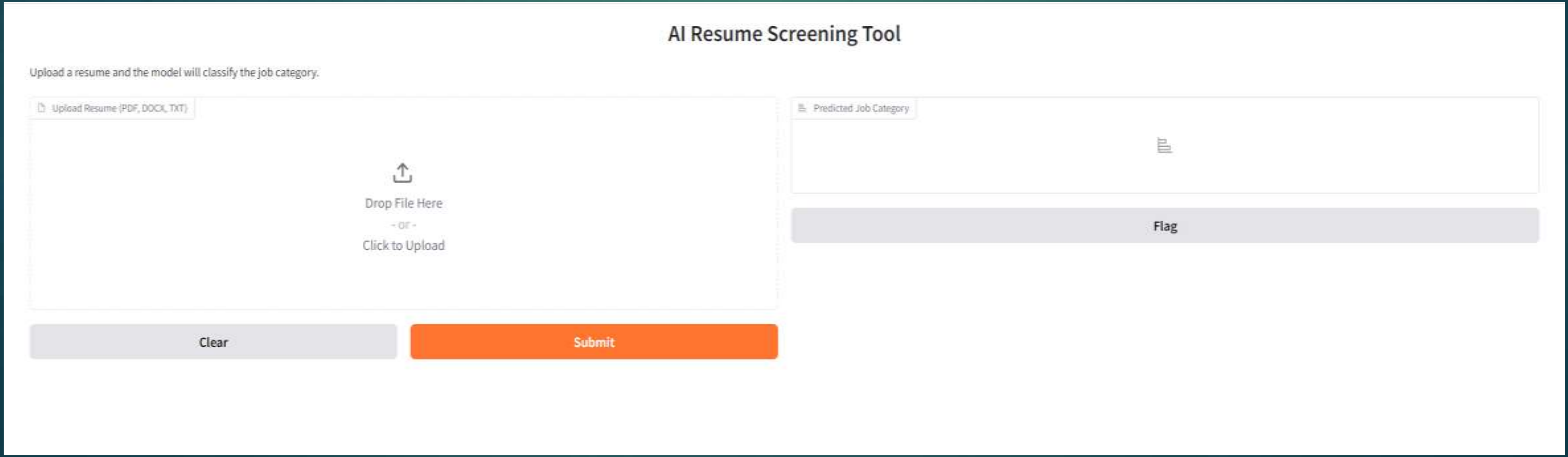
- Originally, we selected SVC because Support Vector Machines are strong classifiers and often perform well on margin-based text classification tasks.
- The alarmingly high accuracy due to improper oversplitting and SVC performance made us change the model
- High-dimensional text requires linear decision boundaries, so we switched to LinearSVC, which is the industry standard for text classification. It is optimized for sparse vectors, significantly faster, and generalizes better for this type of NLP problem.

Linear SVC Benefits

- ✓ Fast
- ✓ Works with TF-IDF
- ✓ More accurate compared to SVC
- ✓ Handles high-dimensional data easily
- ✓ Used in industrial NLP baselines

Resume Screening Tool

- Upload Text/PDFs
- PDFs of resume
- Text to list skills



Resume deployed on: Gradio

WHY GRADIO?

•It runs **100% inside Colab**

•It requires **no local environment**

•It provides a **public web link instantly**

•It supports **file upload** (PDF, DOCX, TXT)

THANK YOU